

Asymptotic Betti Bounds for Local Complete Intersection

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Abstract. In this paper, we show that for any local complete intersection subvariety $Y \subset X$ defined by r global sections of the power of a very ample line bundle $\mathcal{L}^{\otimes d_i}$ with $d_i \leq d$, in a n -dimensional projective variety X with at most local complete intersection singularities, there is a uniform constant C and we have the following asymptotically optimal upper bound of the total Betti number of Y :

$$\deg(X) \binom{n-1}{r-1} d^n + C d^{n-1}.$$

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1 Introduction

1.1. In this paper, a variety is defined to be a separated scheme of finite type over an algebraically closed field. Varieties are not necessarily assumed to be irreducible or reduced as the constructible sheaves do not detect the scheme structure.

We mainly work in the Betti setting for simplicity: all varieties are assumed to be defined over $\mathbb{C}$ and all complexes are taken to be $\mathbb{Q}$ -modules. Nevertheless, all of our results also hold in the étale setting equally well.

The total Betti number of a variety V is denoted by

$$B(V; \mathbb{Q}) = \sum_{i=0}^{2 \dim V} h^i(V; \mathbb{Q}) = \sum_{i=0}^{2 \dim V} \dim H^i(V; \mathbb{Q}).$$

The Euler characteristic of a variety V is denoted by

$$\chi(V; \mathbb{Q}) = \sum_{i=0}^{2 \dim V} (-1)^i h^i(V; \mathbb{Q}).$$

For the compact supported version, the notations are similar.

Bounds for Betti numbers are widely used in Weil-type problems in arithmetic geometry. Let $f_1, \dots, f_r$ be polynomials defined over the finite field $\mathbb{F}_q$, where $q = p^a$ is a power of a prime number

p . Suppose these equations define a k -dimensional affine l.c.i. geometrically irreducible variety $V \subset \mathbb{A}^N$. Then the number of common solutions of the equations over $\mathbb{F}_q$ is precisely the number of $\mathbb{F}_q$ -rational points of V .

The Lang–Weil estimate states that:

$$|\#V(\mathbb{F}_q) - q^k| \leq B_c(V \times_{\mathbb{F}_q} \overline{\mathbb{F}_q}; \mathbb{Q}_\ell) \cdot q^{k-\frac{1}{2}}, \quad (1)$$

where ℓ is a prime number different from p .

Our main theorem, Theorem 3.2, gives an estimate of the Betti number over an algebraically closed field. Hence, combining with the Lang–Weil estimate (1), we can dominate not only the main term but also the error term of the number of $\mathbb{F}_q$ -points.

1.2. Betti number bounds. Let $(X, \mathcal{L})$ denotes our ambient space, where X is an n -dimensional projective variety over an algebraically closed field with at most local complete intersection singularities and $\mathcal{L}$ is a very ample line bundle on X . Hence there is a natural embedding

$$X \hookrightarrow \text{Proj Sym}^\bullet H^0(X, \mathcal{L})^\vee \cong \mathbb{P}^N.$$

Let $Y \subset X$ be a subvariety defined by sections

$$f_i \in H^0(X, \mathcal{L}^{\otimes d_i}), \quad 1 \leq i \leq r,$$

via the Cartesian diagram:

$$\begin{array}{ccc} X & \xhookrightarrow{\quad} & \mathbb{P}^N \\ \uparrow & & \uparrow \\ Y & \hookrightarrow & \{f_1 = f_2 = \cdots = f_r = 0\} \end{array}.$$

We call d_i the degree of the section f_i , and assume that $d_i \leq d$ for all i . If Y is l.c.i., we have

$$\dim Y = n - r.$$

In this paper, we establish bounds for l.c.i. subvarieties in l.c.i. projective variety X :

Theorem 1.1 (Main theorem = Theorem 3.2). *Let X be a dimension n l.c.i. variety, $\mathcal{L}$ be a very ample line bundle on X . Then for all l.c.i. subvarieties $Y \subset X$ defined by r global sections of degree d_i power of $\mathcal{L}$, $d_i \leq d$, we have a uniform and sharp upper bound for the total Betti number of Y :*

$$B(Y; \mathbb{Q}) \leq \deg(X) \binom{n-1}{r-1} d^n + C d^{n-1}. \quad (2)$$

We say that an upper bound is asymptotically sharp if the leading coefficient cannot be improved.

1.3. Historical approaches to Betti bounds. In [14], Milnor established an upper bound for the sum of the Betti numbers of a complex affine algebraic variety V . More precisely, suppose $V \subset \mathbb{C}^N$, $N \geq 1$, is defined by $r \geq 1$ equations F_i of degree at most d . Milnor proved that:

$$B(V; \mathbb{Q}) \leq d(2d-1)^{2N-1}.$$

His proof reduces the problem to a real hypersurface and applies Morse theory to do Chain-level counting of critical points, resulting in the unpleasant exponent $(2N - 1) =$ the real dimension of the real hypersurface, than $\dim X$ in our result.

Subsequently, many nontrivial attempts were made to generalize this type of result to arbitrary fields, since the above approach is too topological. The first strategy is to estimate the Euler characteristic; see, for example, [6] and [2]. One then derives bounds for the total Betti number from bounds on the Euler characteristic by Katz's trick; see [10].

The key idea in Katz's trick is the use of the affine weak Lefschetz theorem to reduce the problem of bounding the total Betti number to that of bounding the middle cohomology. In [19], this idea was further developed using perverse sheaves to estimate l.c.i. subvarieties in $\mathbb{A}^n$ or $\mathbb{P}^n$:

$$B_c(V; \mathbb{Q}_\ell) \leq \binom{n-1}{r-1} d^n + C d^{n-1}. \quad (3)$$

Their key tool is the perverse degeneration lemma in this paper, which was first proved in [11] by Katz.

Furthermore, the smooth assumption on the ambient space or the l.c.i. assumption on the subvarieties can be removed. It's firstly considered in [15] for the codimension-one case. Their method relies on a more technical homological algebra result called the proper degeneration lemma, which introduces additional contributions from non-middle cohomology and yields a bound larger than ours.

1.4. New methods in this paper. Comparing with [19], we overcome the difficulties arising from singular ambient spaces by using Chern–Schwarz–MacPherson classes instead of ordinary Chern classes.

The proof of our main theorem, Theorem 3.2, proceeds by several reductions to generic and affine cases, which we can compute the leading term of its Euler characteristic precisely in Lemma 3.1. Then we'll show that all the reduction steps do not affect the leading term, and our final Betti number will have same leading term as the leading term of the Euler characteristic above.

1.5. Organization. In Chapter 2, we introduce the preliminaries including lemmas on perverse sheaves and computational tools of Chern-Schwarz-MacPherson classes; chapter 3 is devoted to proving the main theorem; in Chapter 4, we introduce an application estimating the Betti number of singular cohomology of real algebraic varieties by Smith-Thom inequality.

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2 Preliminaries

2.1. Perverse sheaves. For the fundamentals of perverse sheaves such as Artin vanishing and proper base change, one can refer to standard textbooks such as [1] and [5]. Here we only present several lemmas we'll use next.

For simplicity, all sheaves mean constructible complexes and all six functors are in the derived sense in this paper. All our notation depending on the perversity refers to the middle perversity.

Definition 2.1 (Set-theoretically local complete intersection). Let V be a variety over k of pure dimension n . We say V is set-theoretically a local complete intersection if there exists an open affine cover $V = \bigcup_{m \in I} U_m$ and with the following property: for each $m \in I$, there exists an integer $r \geq 0$, and polynomials $f_1, \dots, f_r \in k[T_1, \dots, T_{n+r}]$ such that

$$U_{m,\text{red}} \cong \text{Spec}\left(\frac{k[T_1, \dots, T_{n+r}]}{\sqrt{(f_1, \dots, f_r)}}\right).$$

Clearly, any local complete intersection variety is set-theoretically a local complete intersection. We'll call them set theoretically l.c.i. or l.c.i. for simplicity through this paper.

Theorem 2.2 (Perversity of l.c.i.). *Let V be a variety of pure dimension n . If V is set-theoretically a local complete intersection, then $\mathbb{Q}_V[n]$ is a perverse sheaf.*

Proof. Since $\mathbb{Q}_V[n]$ is a shifted local system by its dimension, it is automatically perverse connective. Because perversity is a local property and V is set-theoretically l.c.i., we may assume that V is a hypersurface in $\mathbb{A}^{n+1}$. Set $\mathcal{P} = \mathbb{Q}_{\mathbb{A}^{n+1}}[n+1]$, which is a perverse sheaf as $\mathbb{A}^{n+1}$ is smooth. We have the excision exact triangle:

$$\mathcal{P}[-1] \rightarrow i_* i^* \mathcal{P}[-1] \rightarrow j_! j^* \mathcal{P} \rightarrow, \quad (4)$$

where $i : V \rightarrow \mathbb{A}^{n+1} \leftarrow U : j$, and U is the complement of V .

apply the perverse cohomology functor ${}^p\mathcal{H}$ to (4), then:

$${}^p\mathcal{H}^{k-1}\mathcal{P} \rightarrow {}^p\mathcal{H}^k(i_* i^* \mathcal{P}[-1]) \rightarrow {}^p\mathcal{H}^k j_! j^* \mathcal{P} \rightarrow, \quad (5)$$

By the naturality of pushforward and pullback, the middle term of (5) can be identified with ${}^p\mathcal{H}^k \mathbb{Q}_V[n]$. Since j is an affine open immersion, the functor $j_!$ is perverse t -exact. Therefore both the left and right terms in (5) are perverse sheaves. Thus the middle term ${}^p\mathcal{H}^k \mathbb{Q}_V[n]$ vanishes if $k \neq 0, 1$, i.e. $\mathbb{Q}_V[n]$ is perverse coconnective. $\square$

Theorem 2.3 (Perverse degeneration lemma). *Let C be an affine, geometrically connected curve, and let $\mathcal{P}$ be a perverse coconnective sheaf on C . Let*

$$i : \{s\} \hookrightarrow C \quad \text{and} \quad j : U \hookrightarrow C$$

be the inclusion of a closed point $s \in C$ and its complement $U = C \setminus \{s\}$, respectively. Then there is an injection of $\mathbb{Q}$ -vector spaces:

$$\mathcal{H}^{-1} \mathcal{P}_s \rightarrow i^* R^0 j_* \mathcal{H}^{-1} j^* \mathcal{P}. \quad (6)$$

Proof. We have the excision exact triangle:

$$i_* i^! \mathcal{P} \rightarrow \mathcal{P} \rightarrow j_* j^* \mathcal{P}. \quad (7)$$

Apply i^* to (7), we obtain:

$$i^! \mathcal{P} \rightarrow \mathcal{P}_s \rightarrow i^* j_* j^* \mathcal{P}. \quad (8)$$

Take cohomology to (8) gives the long exact sequence:

$$\mathcal{H}^{-1} i^! \mathcal{P} \rightarrow \mathcal{H}^{-1} \mathcal{P}_s \rightarrow i^* \mathcal{H}^{-1} j_* j^* \mathcal{P}, \quad (9)$$

Since $\mathcal{P}$ is perverse coconnective, the left term of (9) vanishes, thus we have a injection of the right two terms. Therefore, it suffices to prove that

$$\mathcal{H}^{-1} j_* \mathcal{Q} = R^0 j_* \mathcal{H}^{-1} \mathcal{Q}, \quad (10)$$

where $\mathcal{Q} = j^* \mathcal{P}$. To prove (10), we consider the Leray spectral sequence:

$$E_2^{pq} = R^p j_* \mathcal{H}^q \mathcal{Q} \Rightarrow E_\infty^{p+q} = \mathcal{H}^{p+q} j_* \mathcal{Q}, \quad (11)$$

By Artin vanishing, we have $R^p j_* \mathcal{F} = 0$ for $p \geq 1$, and since $\mathcal{Q}[-1]$ is also perverse coconnective, we have $\mathcal{H}^q(\mathcal{Q}) = 0$ for $q \leq -2$. Hence, only the term $(p, q) = (0, -1)$ contributes to the total degree -1 cohomology, proving (10). The E_2 -page degenerates as the following:

$$\begin{array}{ccccccc}
 & & 0 & & & & \\
 & & \swarrow \cdots & & & & \\
 & 0 & & 0 & & E_2^{00} & \\
 & \swarrow & & \swarrow d_2 & & & \\
 & 0 & & 0 & & E_2^{0,-1} & \\
 & \swarrow & & \swarrow & & \searrow d_2 & \\
 & 0 & & 0 & & 0 & \\
 & \swarrow & & \swarrow & & \searrow & \\
 & 0 & & 0 & & 0 & \\
 & & & & & & 0
 \end{array}$$

□

We apply Theorem 2.3 to the case where C is $\mathbb{A}^1$ as the base curve of an algebraic family, and the perverse sheaf is so-called "perverse sheaf that comes from geometry". The key ingredient is the proper base change theorem.

Corollary 2.4. *Let Y be a $(e+1)$ -dimensional set-theoretically l.c.i. variety. Let $\pi : Y \rightarrow \mathbb{A}^1$ be an affine morphism, and denote by Y_0 and Y_η the special and generic fibres, respectively, both of dimension e . Then there is an injection:*

$$H_c^e(Y_0) \hookrightarrow H_c^e(Y_\eta). \quad (12)$$

Consequently, for every geometric point $t \in U = \mathbb{A}^1 \setminus \{0\}$,

$$h_c^e(Y_0) \leq h_c^e(Y_t). \quad (13)$$

Proof. Consider the communicative diagram:

$$\begin{array}{ccc} Y & \xrightarrow{\pi} & \mathbb{A}^1 \\ \iota \uparrow & & \uparrow i \\ Y_0 & \xrightarrow{p} & \{0\} \end{array} \quad (14)$$

By the proper base change theorem,

$$H_c^e(Y_0; \mathbb{Q}) = R^e p_{!} \iota^* \mathbb{Q}_Y = i^* R^e \pi_{!} \mathbb{Q}_Y. \quad (15)$$

Since Y is dimension $e + 1$ set-theoretically l.c.i, by Theorem 2.2, $\mathbb{Q}_Y[e + 1]$ is perverse. And by $\pi_{!}$ is perverse t-exact, we may apply Theorem 2.3 to $\mathcal{P} = \pi_{!} \mathbb{Q}_Y[e + 1]$ on $\mathbb{A}^1$, then we have:

$$\mathcal{H}^{-1} \mathcal{P}_0 = H_c^e(Y_0; \mathbb{Q}). \quad (16)$$

Similarly, applying proper base change to:

$$\begin{array}{ccc} Y & \longrightarrow & \mathbb{A}^1 \\ \uparrow & & \uparrow j \\ Y_\eta & \longrightarrow & U \end{array} \quad (17)$$

gives:

$$i^* R^0 j_* \mathcal{H}^{-1} j^* \mathcal{P} = H_c^e(Y_\eta; \mathbb{Q}). \quad (18)$$

Hence (12) follows. It remains to prove that $h_c^e(Y_t) = h_c^e(Y_\eta)$ for general t . Since $R^e \pi_{!} \mathbb{Q}_Y$ is a constructible sheaf on $\mathbb{A}^1$, there exists a Zariski open dense subset on which the fibre dimension is constant. By proper base change, the fibre of $R^e \pi_{!} \mathbb{Q}_Y$ at a point t is naturally identified with $H_c^e(Y_t)$. Therefore the dimensions agree with that of the generic fibre on a dense open subset, completing the proof. $\square$

Example 2.1. Now we construct an affine family $\tilde{Y} \rightarrow \mathbb{A}^1$ where the special fibre Y_0^0 is given e -dimensional l.c.i. variety; and general fibres $Y_t^0 \cong Y^0$.

Let f_i^0 be the defining equations of Y_0^0 , and let g_i^0 be sufficiently general polynomials of the same degree, defining Y^0 . Define $\tilde{Y} \subset X \times \mathbb{A}^1$ by the equations $(1 - t)f_i^0 + tg_i^0$, $1 \leq i \leq r$. Thus by definition, the total space $\tilde{Y}$ is also set-theoretically l.c.i., thus we can apply (2.4) and we have

$$h_c^e(Y_0) \leq h_c^e(Y_t). \quad (19)$$

Theorem 2.5 (Perverse weak Lefschetz theorem). *Let X be an algebraic variety over an algebraically closed field, and let $\iota : X \rightarrow \mathbb{P}^N$ be a quasi-finite morphism. Suppose $\mathcal{F}$ is perverse connective constructible complex on X . Then, for a general hyperplane H in $\mathbb{P}^N$, the Gysin map $H_c^{i-2}(X \times_{\mathbb{P}^N} H; \mathcal{F}) \rightarrow H_c^i(X; \mathcal{F})$ is surjective for $i = 1$ and is an isomorphism for all $i \geq 2$.*

Proof. It's [18], Lemma 3.8. $\square$

Corollary 2.6 (Affine weak Lefschetz theorem for (singular) varieties). *Let X be a n -dimensional affine algebraic variety in $\mathbb{A}^n$. Suppose $\mathcal{F}$ is perverse connective constructible complex on X . Then, for a general hyperplane H in $\mathbb{A}^n$, the Gysin map $H_c^{i-2}(X \cap H; \mathbb{Q}) \rightarrow H_c^i(X; \mathbb{Q})$ is surjective for $i = n - 1$ and is an isomorphism for all $i \leq n - 2$.*

Proof. It's trivial by applying Theorem 2.5 to the case $\iota : X \rightarrow \mathbb{A}^n$ and $\mathcal{F} = \mathbb{Q}_X[n]$. $\square$

Compare with the smooth version in [10], it's up to a Poincare dual. Although in singular case we can't recover his version, it's enough to estimate the dimension of the cohomology groups.

2.2. Chern-Schwarz-MacPherson class Chern-Schwarz-MacPherson class is a replacement for usual Chern class on singular spaces, which was first proposed by Grothendieck in his autobiography [9], Note 87₁.

Let $\mathbf{D}_c^b(X)$ denote the bounded derived category of constructible complexes on variety X ; **SmProj** denote the category of smooth projective varieties with smooth proper morphisms; **Proj** denote the category of projective varieties with proper morphisms; **Ab** denote the category of abelian groups.

Theorem 2.7 (Existence of Chern-Schwarz-MacPherson classes over $\mathbb{C}$). *Let the two functors $CF, H_\bullet : \mathbf{Proj} \rightarrow \mathbf{Ab}$ send a projective variety X over $\mathbb{C}$ to the group of constructible functions on X and the homology group of X respectively. Restricted to **SmProj**, we have a natural transformation: for X smooth projective over $\mathbb{C}$, we have the group homomorphism sending the basis elements as following:*

$$CF(X) \rightarrow H_\bullet(X) \quad (20)$$

$$\mathbf{1}_V \mapsto c(TV) \cap [V] \quad (21)$$

Then our assertion is that we can uniquely extend this natural transformation to the two functors defined on **Proj**. More precisely, there exists $c_{SM} : CF(X) \rightarrow H_\bullet(X)$ for X projective, satisfying:

- (Naturality for proper pushforward) For $f : X \rightarrow Y$, we have:

$$f_* c_{SM}(\varphi) = c_{SM}(f_* \varphi); \quad (22)$$

- (Linearity)

$$c_{SM}(\varphi_1 + \varphi_2) = c_{SM}(\varphi_1) + c_{SM}(\varphi_2); \quad (23)$$

- (Normalization) If X is smooth, then:

$$c_{SM}(\mathbf{1}_X) = c(TX) \cap [X]; \quad (24)$$

- (Uniqueness) The homomorphism c_{SM} satisfying the upper axioms (22), (23) and (24) is unique.

We call the $c_{SM}(\mathbf{1}_X)$ the Chern-Schwarz-MacPherson class and denote it as $c_{SM}(X)$ for simplicity.

Proof. It's [13], Theorem 1. We briefly recall its proof.

Firstly, the construction is two steps: first defining the local Euler obstruction to define a isomorphism between $Z(X)$ and $CF(X)$, then using the so-called Chern-Mather class of a cycle to

define it on singular variety.

Secondly, the most nontrivial part in his proof is the naturality property (22). It may be viewed as a discrete analog of the Grothendieck–Riemann–Roch theorem, since constructible functions play the role of a discrete version of coherent sheaves. The key construction of the proof is the famous MacPherson’s graph construction, which is also analogous to the proof of Grothendieck–Riemann–Roch. $\square$

Remark 2.1. A subtle point is that, in positive characteristic, extending the natural transformation to arbitrary proper morphisms might fail; see [16], Example 6.10. Nevertheless, the following existence theorem, Theorem 2.8, restricts to smooth objects at the level proved in [16], Section 6 via microlocal geometry, which works in arbitrary characteristic.

For constructible complexes rather than constructible functions, one uses Grothendieck’s sheaf–function correspondence:

$$K_0(\mathbf{D}_c^b(X)) \cong CF(X).$$

Hence, there is no nontrivial issue and the corresponding definitions and statements are analogous:

Theorem 2.8 (Chern-Schwarz-MacPherson classes over arbitrary fields). *Let X be a smooth variety. There is a group homomorphism $c_{SM} : K_0(\mathbf{D}_c^b(X)) \rightarrow CH_\bullet(X)$, which assigns to each constructible complex $\mathcal{F}$ a characteristic class $c_{SM}(\mathcal{F})$ in the Chow group.*

Proof. It’s [16], Coro 6.9 (1), where he also proved the naturality with respect to the smooth pullback.

We briefly recall the construction using microlocal geometry, it’s the Segre class of the characteristic cycle on the cotangent bundle.

Assume X is of pure dimension n . The cotangent bundle $T^*X = \text{Spec}_X \text{Sym}^\bullet(\Omega_{X/k}^1)^\vee$ admits a natural compactification by adding a hyperplane at infinity:

$$T^*X \hookrightarrow \mathbb{P}(T^*X \times_X \mathbb{A}_X^1) = \text{Proj}_X \text{Sym}^\bullet(\Omega_{X/k}^1 \oplus \mathcal{O}_X)^\vee.$$

Here, $\mathbb{P}(T^*X \times_X \mathbb{A}_X^1) = \mathbb{P}(T^*X) \amalg T^*X$, and $\mathbb{P}(T^*X)$ plays the role of this hyperplane at infinity. Given a constructible complex $\mathcal{F}$ on X , denote its characteristic cycle by $CC(\mathcal{F}) \in CH_n(T^*X)$ is the cycle supported on the singular support and satisfying the Milnor formula, see [16], Definition 5.10. Its projectivization $\overline{CC}(\mathcal{F}) = \mathbb{P}(CC(\mathcal{F}) \times_X \mathbb{A}_X^1) \in CH_n(\mathbb{P}(T^*X \times_X \mathbb{A}_X^1))$. The characteristic class $cc_X(\mathcal{F})$ is then defined via the following formula of Segre class:

$$cc_X(\mathcal{F}) = c(\Omega_X^1) \cap \pi_* \left(\sum_{i \geq 0} c_1(\mathcal{O}(1))^i \cap \overline{CC}(\mathcal{F}) \right) \in CH_*(X),$$

where $c(\Omega_X^1)$ is the total Chern class of the cotangent bundle, $\pi : \mathbb{P}(T^*X \times_X \mathbb{A}_X^1) \rightarrow X$ denotes the projection, and $\mathcal{O}(1)$ is the dual of the relative tautological bundle. $\square$

For more interesting story on its microlocal aspects, see [8] and [16].

Remark 2.2. In characteristic zero, the two definitions coincide up to a sign:

$$c_{\text{MacPherson},i}(\mathcal{F}) = (-1)^i c_{\text{Saito},i}(\mathcal{F}). \quad (25)$$

Thus in the following Theorems, the result is also up to a sign than the original result in [3] and [4], as in this paper, we use Saito's sign convention.

Remark 2.3. Note that in Theorem 2.7, we use the homology group, while in Theorem 2.8, we use the Chow group. This is because although MacPherson's proof [13] is for homology group only, Kennedy [12] showed that it factors through the Chow group.

Corollary 2.9 (Index theorem). *Let X be a smooth proper variety. Then for any constructible complex $\mathcal{F}$, we have:*

$$\chi(X; \mathcal{F}) = \int_X c_{SM}(\mathcal{F}). \quad (26)$$

Proof. Apply the naturality to the structure morphism $X \rightarrow \text{Spec}(k)$, which is smooth proper. $\square$

We now present the main computational tool in this paper:

Theorem 2.10 (Chern-Schwarz-MacPherson classes of any hypersurfaces). *Let X be a hypersurface in a nonsingular complete variety V . Then*

$$c_{SM}(\mathbf{1}_X) = c(TV) \cap \frac{[X]}{1 + [X]} + \mathcal{M}(X, V) \quad (27)$$

*in A_*V , where the second term is the so-called Milnor class.*

Proof. It's [3], Theorem I.4. $\square$

The first term in (27) is called the Fulton-Johnson class. Roughly speaking, the first term describes the generic contribution, whereas the second term records the contribution coming from singularities. In this paper, we'll avoid direct computations of Milnor classes, since they are usually difficult to compute explicitly. Instead, we reduce to the general-position case using the perverse degeneration lemma, Theorem 2.3. The following statement shows that, in general case, only the Fulton-Johnson class survives:

Corollary 2.11 (Chern-Schwarz-MacPherson classes of general hypersurfaces). *Let $U \subset \mathbb{P}^n$ be any locally closed set (so that $\mathbf{1}_U$ is a constructible function). Then for a general hyperplane $H \subset \mathbb{P}^n$,*

$$c_{SM}(\mathbf{1}_{U \cap H}) = \frac{[H]}{1 + [H]} c_{SM}(\mathbf{1}_U) \quad (28)$$

*in A_*V .*

Proof. It's [4], Prop 2.6, it's obvious to prove it for U and $U \cap H$ smooth and by usual Chern classes. $\square$

For constructible complexes, we have:

Corollary 2.12. *Let X be a smooth projective variety with a very ample invertible sheaf $\mathcal{L}$. Let $\mathcal{F}$ is a complex of constructible sheaves. Let $\iota : Y \rightarrow X$ be the inclusion morphism of the hypersurface Y defined by a general section of $\mathcal{L}^{\otimes d}$. Then we have:*

$$c_{SM}(\iota_* \iota^* \mathcal{F}) = \frac{dc_1(\mathcal{L})}{1 + dc_1(\mathcal{L})} \cap c_{SM}(\mathcal{F}). \quad (29)$$

Proof. This is [15], Lemma 3.3.6. □

Iterating this result yields the following l.c.i. version:

Corollary 2.13 (Chern-Schwarz-MacPherson classes of l.c.i.). *Let $\iota : Y \rightarrow X$ be the inclusion morphism of the l.c.i. subvariety Y defined by r general section of $\mathcal{L}^{\otimes d_i}$, $1 \leq i \leq r$. Then we have:*

$$c_{SM}(\iota_* \iota^* \mathcal{F}) = \prod_{i=1}^r \frac{d_i c_1(\mathcal{L})}{1 + d_i c_1(\mathcal{L})} \cap c_{SM}(\mathcal{F}). \quad (30)$$

Although the lower-degree terms of $c_{SM}(\mathcal{F})$ are generally difficult to compute, the top-dimensional term is explicit, which is sufficient for our computation of the leading term:

Proposition 2.14. *Let X be a smooth variety. Let $\mathcal{F}$ be a constructible complex on X . Let S be the support of $\mathcal{F}$ endowed with the reduced structure. Let $n = \dim S$. Then we have:*

$$c_{SM,i}(\mathcal{F}) = \begin{cases} 0; & j > n, \\ (-1)^n \text{rank}^{vir}(\mathcal{F}) [S]_n; & j = n, \end{cases} \quad (31)$$

where $[S]_n$ is the fundamental class of the union of n -dimensional irreducible components of S , $\text{rank}^{vir}(\mathcal{F}) = \sum_{j \in \mathbb{Z}} (-1)^j \text{rank}_{\mathbb{Q}}(\mathcal{H}^j \mathcal{F})$ is called the virtual rank of $\mathcal{F}$, j pullback $\mathcal{F}$ to the largest purely n -dimensional Zariski open dense subset of S over which $\mathcal{H}^i \mathcal{F}$ is a $\mathbb{Q}$ -local system.

Proof. It's [16], Lemma 6.9(2) or [15], Lemma 3.3.9. □

Example 2.2. Let X is a n -dimensional irreducible projective variety. Let $\mathcal{F} = i_* \mathbb{Q}_X[n]$, where $i : X \rightarrow \mathbb{P}^{\mathbb{N}}$ is an inclusion morphism. Then its support $S = X$ and $\text{rank}^{vir}(\mathcal{F}) = 1$ as it's a local system, thus $c_{SM,n}(\mathcal{F}) = (-1)^n [X]$.

3 Proof of the Main Theorem

We begin with the following computation in the generic case:

Lemma 3.1 (Euler characteristic of general subvarieties). *Let X be a dimension n l.c.i. variety, $\mathcal{L}$ be a very ample line bundle on X . Then for all subvarieties $Y \subset X$ defined by r **general** global sections of degree d_i power of $\mathcal{L}$, $d_i \leq d$, we have a uniform constant C_1 , such that there is an asymptotically sharp upper bound of the absolute value of the Euler characteristic of Y :*

$$|\chi(Y)| \leq \deg(X) \binom{n-1}{r-1} d^n + C_1 d^{n-1} \quad (32)$$

Proof. Consider inclusion morphisms:

$$\begin{array}{c} Y \longrightarrow X \xrightarrow{i} \mathbb{P}^N \\ \quad \quad \quad \searrow \quad \quad \nearrow \\ \quad \quad \quad \iota \end{array}, \quad (33)$$

by functoriality, we have:

$$\chi(Y) = \chi(Y; \iota^* i_* \mathbb{Q}_X) = \chi(\mathbb{P}^N; \iota_* \iota^* i_* \mathbb{Q}_X), \quad (34)$$

then we apply (26) and (30), we have:

$$\chi(\mathbb{P}^N; \iota_* \iota^* i_* \mathbb{Q}_X) = \int_{\mathbb{P}^N} c_{SM}(\iota_* \iota^* i_* \mathbb{Q}_X) = \int_{\mathbb{P}^N} \prod_{i=1}^r \frac{d_i c_1(\mathcal{L})}{1 + d_i c_1(\mathcal{L})} \cap c_{SM}(i_* \mathbb{Q}_X). \quad (35)$$

Now we can apply Example 2.2, thus the top-degree contribution is $[X]$, up to a sign; while all lower-dimensional components contribute only lower-order terms, thus we obtain:

$$|\chi(Y)| \leq \left| \int_{\mathbb{P}^N} \prod_{i=1}^r \frac{d_i c_1(\mathcal{L})}{1 + d_i c_1(\mathcal{L})} \cap [X] \right| + C_1 d^{n-1}. \quad (36)$$

By definition of the degree of a projective variety:

$$\int_{\mathbb{P}^N} c_1(\mathcal{L})^n \cap [X] = \deg X, \quad (37)$$

By (36) and (37), applying $d_i \leq d$, we have:

$$|\chi(Y)| \leq (\deg X) \sum_{i_1 + \dots + i_r = n, i_j \geq 1 \text{ for all } j} d_1^{i_1} \dots d_r^{i_r} + C_1 d^{n-1} \quad (38)$$

$$\leq (\deg X) \left(\sum_{i_1 + \dots + i_r = n, i_j \geq 1 \text{ for all } j} 1 \right) d^n + C_1 d^{n-1}, \quad (39)$$

The number of such compositions equals the number of partitions of $(n - r)$ into r nonnegative parts, which equals $\binom{n-1}{r-1}$ by the “stars and bars” counting. Hence, we complete our proof. $\square$

Theorem 3.2 (Main theorem). *Let X be a dimension n l.c.i. variety, $\mathcal{L}$ be a very ample line bundle on X . Then for all l.c.i. subvarieties $Y \subset X$ defined by r global sections of degree d_i power of $\mathcal{L}$, $d_i \leq d$, we have a uniform constant C , such that there is a asymptotically sharp upper bound of the total Betti number of Y :*

$$B(Y) \leq \deg(X) \binom{n-1}{r-1} d^n + C d^{n-1}. \quad (40)$$

Proof. Assume Y is general. Write $Y = Y^0 \amalg Y^1$, where $Y^1 = Y \cap H$, H is a general hyperplane in $\mathbb{P}^N$, i.e. defined by a general linear section of $\mathcal{L}$. Then Y^0 is affine, while Y^1 is projective of dimension one less. Since we have $Y^1 \hookrightarrow X \cap H$, where $X \cap H$ is of dimension $n - 1$, and we can run induction on the dimension of the ambient space n .

From the excision long exact sequence, we have $\chi(Y) = \chi_c(Y^0) + \chi(Y^1)$. Thus Lemma 3.1 implies:

$$|\chi_c(Y^0)| \leq |\chi(Y)| + |\chi(Y^1)| \quad (41)$$

$$\leq \deg(X) \binom{n-1}{r-1} d^n + (C_1 + \binom{n-2}{r-1}) d^{n-1} \quad (42)$$

$$= \deg(X) \binom{n-1}{r-1} d^n + C_2 d^{n-1}. \quad (43)$$

Consider middle cohomology $h_c^{n-r}(Y^0) = (-1)^{n-r}\chi(Y^0) + h_c^{n-r+1}(Y^0) - h_c^{n-r+2}(Y^0) + \dots$. Because $\mathbb{Q}_Y[n-r]$ is perverse connective, we can apply the affine weak Lefschetz, Corollary 2.6, to the non-middle terms:

$$h_c^{n-r}(Y^0) \leq (-1)^{n-r}\chi(Y^0) + h_c^{n-r-1}(Y^0 \cap H') - h_c^{n-r}(Y^0 \cap H') + \dots \quad (44)$$

$$= (-1)^{n-r}\chi(Y^0) + (-1)^{n-r-1}\chi(Y^0 \cap H'), \quad (45)$$

as only the first non-middle term $h_c^{n-r+1}(Y^0) \leq h_c^{n-r-1}(Y^0 \cap H')$, others are equal.

By induction on n again:

$$h_c^{n-r}(Y^0) \leq |\chi(Y^0)| + |\chi(Y^0 \cap H')| = |\chi(Y^0)| + C_3 d^{n-1}, \quad (46)$$

Combining with (41), we obtain:

$$h_c^{n-r}(Y^0) \leq \deg(X) \binom{n-1}{r-1} d^n + (C_2 + C_3) d^{n-1}. \quad (47)$$

Now we construct a family $\tilde{Y} \rightarrow \mathbb{A}^1$ s.t. the general fibre $Y_t^0 = Y^0$ considered above, while the special fibre Y_0^0 is the affine part of original variety Y_0 . Thus our aim is to prove the formula (40) for Y_0 .

Such a family has been constructed in Example 2.1 with f_i^0 being the dehomogenizations of the defining equations of Y_0 and g_i^0 being sufficiently general polynomials of the same degree.

Apply perverse degeneration lemma, Corollary 2.4 to this family as in Example 2.1, combining with (47), we obtain:

$$h_c^{n-r}(Y_0^0) \leq h_c^{n-r}(Y_t^0) \leq \deg(X) \binom{n-1}{r-1} d^n + (C_2 + C_3) d^{n-1}, \quad (48)$$

We apply affine weak Lefschetz lemma Corollary 2.6 and run induction again, combining with (48):

$$B_c(Y_0^0) \leq h_c^{n-r}(Y_0^0) + \sum_{n-r+1 \leq i \leq 2(n-r)} h_c^i(Y_0^0) \quad (49)$$

$$\leq h_c^{n-r}(Y_0^0) + \sum_{n-r-1 \leq i \leq 2(n-r-1)} h_c^i(Y_0^0 \cap H'') \quad (50)$$

$$\leq h_c^{n-r}(Y_0^0) + C_4 d^{n-1} \quad (51)$$

$$\leq \deg(X) \binom{n-1}{r-1} d^n + (C_2 + C_3 + C_4) d^{n-1}, \quad (52)$$

Finally, we return to the projective variety Y_0 . Write $Y_0 = Y_0^0 \amalg Y_0^1$ and $Y_0^1 = Y_0 \cap H'''$, where H''' is a general hypersurface. Hence the excision long sequence implies $h^i(Y_0) \leq h_c^i(Y_0^0) + h^i(Y_0^1)$. Summing over all i , we have:

$$B(Y_0) \leq B(Y_0^0) + B(Y_0^1). \quad (53)$$

Run induction on n again, we conclude that:

$$B(Y_0) \leq \deg(X) \binom{n-1}{r-1} d^n + C d^{n-1}, \quad (54)$$

thus we complete our proof, which is obviously uniform for all Y . And our result is asymptotically optimal as the reduction step doesn't change the leading term and for the general computation it's exact. $\square$

Remark 3.1. Although we do not explicitly use Katz's original theorem from [10], (44) is essentially the key step of his proof.

Remark 3.2. One can calculate the constant C explicitly by keeping track of the induction steps. For example, see [15], 5.12-5.14.

4 An Application to Topology

An application concerns real algebraic geometry via the Smith-Thom inequality, we can bound the Betti number of singular cohomology of real varieties:

Theorem 4.1 (Smith-thom inequality). *Let X be a finite dimensional CW complex with a cyclic group G -action, we have*

$$\dim_{\mathbb{F}_2} H^i(X^G; \mathbb{F}_2) \leq \dim_{\mathbb{F}_2} H^i(X; \mathbb{F}_2), \quad (55)$$

Proof. It's first proposed in [17]. It can be proven by equivariant cohomology theory; see [7]. $\square$

In particular, let $G = \mathbb{Z}/2\mathbb{Z}$ acting on $\mathbb{C}$ -variety X by complex conjugation, then $X(\mathbb{R})$ is nothing but the fixed point of G . Thus,

$$B(X(\mathbb{R}); \mathbb{F}_2) \leq B(X; \mathbb{F}_2). \quad (56)$$

Therefore, if one can bound the $\mathbb{F}_2$ -Betti numbers of a complex algebraic variety, then one immediately obtains bounds for the topology of its real locus:

Theorem 4.2. *Let Y is a projective, l.c.i. algebraic subvariety in X as in Theorem 3.2, then:*

$$B(X(\mathbb{R}); \mathbb{Z}) \leq \deg(X) \binom{n-1}{r-1} d^n + C d^{n-1}. \quad (57)$$

Proof. The proof of Theorem 3.2 works equally well with coefficients in any local system. So we obtain:

$$B(X; \mathbb{F}_2) \leq \deg(X) \binom{n-1}{r-1} d^n + C d^{n-1}. \quad (58)$$

Combining it with Smith-Thom inequality (56), we have:

$$B(X(\mathbb{R}); \mathbb{F}_2) \leq B(X; \mathbb{F}_2) \leq \deg(X) \binom{n-1}{r-1} d^n + C d^{n-1}, \quad (59)$$

by the universal coefficient theorem, there is a split exact sequence:

$$0 \rightarrow H^i(X(\mathbb{R}); \mathbb{Z}) \otimes \mathbb{F}_2 \rightarrow H^i(X(\mathbb{R}); \mathbb{F}_2) \rightarrow \text{Tor}_1^{\mathbb{Z}}(H^{i+1}(X(\mathbb{R}); \mathbb{Z}), \mathbb{F}_2) \rightarrow 0. \quad (60)$$

Therefore,

$$\dim_{\mathbb{F}_2} H^i(X(\mathbb{R}); \mathbb{F}_2) \geq \dim_{\mathbb{F}_2} (H^i(X(\mathbb{R}); \mathbb{Z}) \otimes \mathbb{F}_2), \quad (61)$$

Then we cancel the torsion: write $H^i(X(\mathbb{R}); \mathbb{Z}) \cong \mathbb{Z}^{b_i} \oplus T_i$, where T_i is the torsion subgroup. Then:

$$H^i(X(\mathbb{R}); \mathbb{Z}) \otimes \mathbb{F}_2 \cong \mathbb{F}_2^{b_i} \oplus (T_i \otimes \mathbb{F}_2). \quad (62)$$

Hence,

$$\dim_{\mathbb{F}_2} (H^i(X(\mathbb{R}); \mathbb{Z}) \otimes \mathbb{F}_2) \geq b_i = \text{rank}_{\mathbb{Z}} H^i(X(\mathbb{R}); \mathbb{Z}). \quad (63)$$

By (59), (61) and (63), we complete our proof. $\square$

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